

# Module 06: Learning — Trial and Error

## Learning Objectives

1. Understand that **Learning** happens when we make mistakes.
2. Learn about **Trial and Error** (Try, Fail, Fix, Repeat).
3. Know that being "bad" at something is the first step to being "good" at it.

## Introduction: The Video Game

Imagine playing a video game. You run to the right -> You fall in a pit. Game Over. Do you quit forever? No! You try again, but this time you JUMP. That is learning.

- **Trial:** Run right.
- **Error:** Fall in pit.
- **Update:** Next time, jump!

## Key Concepts

### 1. The Error Signal (The Surprise)

When you mess up, your brain wakes up. "Whoops! That didn't work!" This feeling is called **Surprise** (or Prediction Error). It feels yucky, but it is useful.

### 2. The Update (The Fix)

You have to change your plan.

- "Too heavy? Use two hands."
- "Too hot? Blow on it." The definition of Insanity is doing the same thing and expecting a different result. The definition of Learning is doing something DIFFERENT.

### 3. Neuroplasticity (The Brain Road)

Every time you practice, the road in your brain gets smoother.

- First time: Bumpy dirt road (Hard).
- 100th time: Super highway (Easy).

## Activities

### Activity 1: The Card Tower

Build a tower with playing cards. It WILL fall. When it falls, say "Great Data!" instead of "Oh no!" Try a different shape. See how high you can get.

### Activity 2: The Blind Drawing

Close your eyes. Try to draw a cat. Open your eyes. It looks funny! Now look at the paper and draw a cat. Your eyes give you **Feedback** to correct your hand.

## Summary

Mistakes are not bad. Mistakes are data. If you aren't failing, you aren't learning!

## References

- *The Girl Who Never Made Mistakes* by Mark Pett
- *Beautiful Oops!* by Barney Saltzberg